

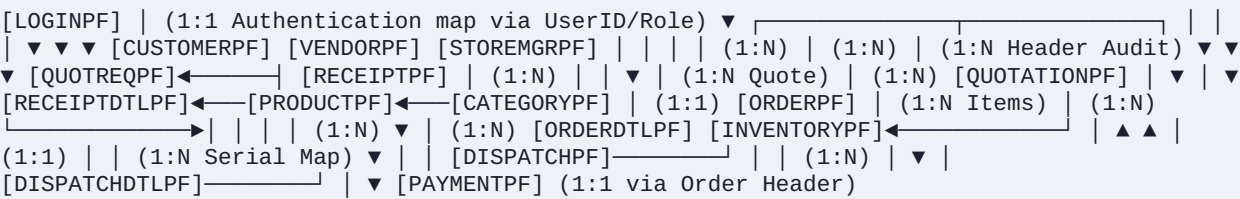
IBM I SUPPLY CHAIN ARCHITECTURE

Relational Database Lifecycle & Physical File (PF) Architecture Specification

1. Entity-Relationship Diagram (ERD)

The following structural map models how data flows across physical files (DB2 for i) from initial vendor acquisition to final dispatch and nightly batch monitoring.

SYSTEM ENTITY-RELATIONSHIP BLUEPRINT



2. Table Relationship & Structural Mapping

To preserve transactional integrity, physical files (PF) link together using strict Logical Keys. This architectural mapping dictates how relationships are governed inside DB2 for i:

Source PF	Target PF	Cardinality	Key Linkage	Operational Purpose
LOGINPF	CUSTOMER / VENDOR / STOREMGR	1 : 0..1 (Cond.)	UserID	Separates credentials from business profile details.
CUSTOMERPF	QUOTREQPF	1 : N	CustomerID	Tracks which customer initiated the purchase inquiry.
QUOTREQPF	QUOTATIONPF	1 : N	RequestID	Allows flexible vendor revision iterations per request.
QUOTATIONPF	ORDERPF	1 : 1	QuotationID	Locks agreed quotes directly into binding customer orders.
ORDERPF	ORDERDTLPF	1 : N	OrderID	Header-to-Detail Pattern: Handles multiple line items.
VENDORPF	RECEIPTPF	1 : N	VendorID	Tracks the origin source of bulk incoming warehouse freight.
STOREMGRPF	RECEIPTPF	1 : N	StoreMgrID	Audit trail identifying the manager responsible for intake.
RECEIPTPF	RECEIPTDTLPF	1 : N	ReceiptID	Header-to-Detail Pattern: Breaks down a bulk delivery box.
PRODUCTPF	INVENTORYPF	1 : N	ProductID	Maps a generic item model to unique tracked units.
LOCATIONPF	INVENTORYPF	1 : N	LocationID	Tracks the exact Rack/Shelf placement inside a warehouse.
ORDERPF	DISPATCHPF	1 : 1	OrderID	Generates a unique delivery manifest and tracking ID.
DISPATCHPF	DISPATCHDTLPF	1 : N	DispatchID	Header-to-Detail Pattern: Lists all unique items in a box.
INVENTORYPF	DISPATCHDTLPF	1 : N	SerialNo	Links outbound items explicitly to serial records.
ORDERPF	PAYMENTPF	1 : 1	OrderID	Ties cash flows to completed customer demands.

3. The 12-Phase Transactional Lifecycle

Below is the complete end-to-end execution path utilizing a concrete real-world case: **TechSource Electronics Pvt Ltd** (V001) supplying Dell Latitude Laptops to **XYZ Retail Solutions** (C001), managed by **Rajat Singh** (SM001).

PHASE 1: SECURE AUTHENTICATION

Impacted Physical File: **LOGINPF**

All users authenticate through a centralized gateway. This file maps **UserID** and encrypted credentials to clear roles (**CUSTOMER** , **VENDOR** , **STORE_MANAGER**) to enforce Role-Based Access Control (RBAC) across applications.

PHASE 2: GOODS RECEIPT & AUDIT INTAKE

Impacted Physical Files: **STOREMGRPF**, **RECEIPTPF**, **RECEIPTDTLPF**

A vendor delivery truck carrying 40 Dell Laptops arrives. The transaction is captured linearly:

- **STOREMGRPF**: Asserts the identity of the physical receiver (**SM001**) for absolute corporate governance.
- **RECEIPTPF**: Logs header info, tying receipt transaction **R001** to Vendor **V001** .
- **RECEIPTDTLPF**: Instantiates granular line entries for each item, assigning distinct serial IDs (**DL001** , **DL002** , etc.).

PHASE 3: PHYSICAL LOCATION ALLOCATION

Impacted Physical Files: **LOCATIONPF**, **INVENTORYPF**

The Store Manager directs inventory to specific real estate. **LOCATIONPF** tracks exact coordinates (Warehouse **WH01** , Rack **A** , Shelf **01** $\rightarrow$ **LOC001**). Simultaneously, records are inserted into **INVENTORYPF** with an initial state status of **IN_STOCK** .

PHASE 4: CUSTOMER REQUEST FOR QUOTATION

Impacted Physical Files: **CUSTOMERPF**, **QUOTREQPF**

Customer **C001** logs in seeking 5 Dell Laptops. This does not impact current stock. It creates a record in **QUOTREQPF** (**QR001**) to formally signal an intention to buy, prompting pricing calculations without holding inventory.

PHASE 5: VENDOR REVIEW & BIDDING

Impacted Physical Files: **VENDORPF**, **QUOTATIONPF**

Vendor **V001** assesses the request against availability matrices. The system checks active warehouse lines (Available = 40, Requested = 5). Upon confirmation, **QUOTATIONPF** emits an official binding offer (**QT001**) priced at **₹250,000** with an **APPROVED** state flag.

PHASE 6: CUSTOMER ORDER COMMIT

Impacted Physical Files: **ORDERPF**, **ORDERDTLPF**

The customer formally signs off on the quote. The system generates an order header in **ORDERPF** (**ORD001** , Status = **PENDING**) containing structural shipping vectors. The breakdown of specific target product lines is isolated in **ORDERDTLPF** .

PHASE 7: STORE MANAGER ORDER VALIDATION

Impacted Physical File: **INVENTORYPF**

The open order hits the fulfillment center. The manager issues a targeted query against **INVENTORYPF** to verify immediate physical availability. Finding sufficient matching stock items, the transaction switches status safely to **APPROVED** .

PHASE 8: INVENTORY STATE RESERVATION

Impacted Physical File: **INVENTORYPF**

To prevent multi-channel double-selling, the system updates five specific serial entries (**DL001** through **DL005**) in **INVENTORYPF** . Their state transitions instantly: **IN_STOCK** → **RESERVED** . This ensures no other customer can claim these physical units.

PHASE 9: PHYSICAL DISPATCH COMPILATION

Impacted Physical Files: **DISPATCHPF**, **DISPATCHDTLPF**

Items are physically picked from **LOC001** . A shipping manifest header is recorded in **DISPATCHPF** including global courier reference routing codes (**BD123456789**). The precise outbound serial pairings are committed explicitly into **DISPATCHDTLPF** .

PHASE 10: FINAL STOCK LEDGER RECONCILIATION

Impacted Physical File: **INVENTORYPF**

As the items leave the loading bay, the system triggers state changes in **INVENTORYPF** , updating the serial statuses: **RESERVED** → **DISPATCHED** . The math equations recalculate real-time system counts: $40 - 5 = 35$ units remaining.

PHASE 11: REVENUE & FINANCIAL SETTLEMENT

Impacted Physical File: **PAYMENTPF**

The customer settles accounts via NEFT transfer. **PAYMENTPF** receives a new transactional entry mapping back directly to **ORD001** , archiving the financial clearing status alongside routing references for accounting reconciliation loops.

PHASE 12: NIGHTLY IBM I BATCH EXCEPTION TRACKING

Impacted Physical Files: **PRODUCTPF**, **INVENTORYPF**, **LOWSTOCKPF**

During low-activity windows, a nightly batch job is submitted via **SBMJOB**. An optimization query compares product alert levels in **PRODUCTPF** against the current live count in **INVENTORYPF**. If a condition matches *CurrentStock* < *ReorderLevel*, a structural record logs into **LOWSTOCKPF**, outputting a printer file (**LOWSTK.PRTF**) accessible via **WRKSPLF**.

4. Architectural System Insights

- **The Inventory Hub Strategy:** The **INVENTORYPF** file is the operational core. Instead of copying large historical data chunks between multiple tracking tables, the ledger uses optimized field state mutations via the asset's **SerialNo**.
- **Decoupled Reporting:** The **LOWSTOCKPF** table avoids heavy live triggers. It leverages the native asynchronous batch capabilities of IBM i, removing execution load from day-time transaction processes.